

KRISHNA DIPAYAN BHUNIA

Senior Data Scientist / Engineer | Deep Learning Engineer | Gen AI Engineer

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SUMMARY

- **Seasoned Senior Data Engineer / Scientist with 12+ years of experience** delivering robust solutions across diverse industries, including CRM, Oil & Gas, Financial, and FMCG. Proven expertise in **Python (6+ years)** as a **Data Engineer/Scientist**, specializing in **data engineering, data analysis, and visualization**.
- Adept at writing clean, production-level code using **Object-Oriented Programming in Python** and highly skilled in **DSA (Data Structure and Algorithm)**, and relational databases. Strong problem-solving and analytical skills with the ability to understand problem domains and analyze datasets.
- Worked as an **Application Developer (Python and C#)**, designing and implementing **Software solutions** with expertise in **back-end technologies**.
- Proficient in **Python**, utilizing libraries like **NumPy, and Pandas** for advanced data analysis and engineering stuff. Skilled in architecting scalable solutions using cloud platforms (**AWS, Azure**) and containerization (**Docker**). Strong background in data engineering processes, algorithm analysis, and database management (SQL, No-SQL). Proven ability to communicate technical insights effectively and collaborate within cross-functional teams.

EDUCATION

- **M.TECH (Computer Eng. with Specialization in Software Eng.)** Aug 2016 – Jul 2018
Veer mata Jijabai Technological Institute (VJTI), Mumbai
- **B.E Computer Engineering (University of Mumbai)** Aug 2008 – Jun 2012

TECHNICAL SKILLS

- **Python, Java, C# (C-Sharp), Docker, GitHub**, Data Science and Data engineering processes, Data Analysis, Classification, Regression Analysis, Cluster Analysis, Data Visualization
- **MS SQL Server (2019/2017), MS SQL, PL-SQL**, Oracle, MySQL, **MongoDB (NO-SQL DB)**, Oracle 19c, **Neo4j (Graph DB)**
- **Machine Learning, Deep-Learning, NumPy, Pandas, Pytorch, Pyspark, Tensorflow, NLP (Natural Language Processing), Statistics**
- **DSA (Data Structures & Algorithms)**, Algorithm Analysis (Space & Time Complexity), Distributed Systems & Architecture
- **AWS (Amazon Web Services), Azure**

WORK EXPERIENCE & PROJECTS

Lead Assistant Vice President - Decision Science in HSBC

May 2025 - Present

Senior Data Scientist in Capgemini

Oct 2021 – Feb 2025

Client - Novartis - (Role - GenAI NLP Engineer/ Data Scientist) -

Remote

- Developed and optimized a **RAG (Retrieval-Augmented Generation)** pipeline for processing medical drug manufacturing documents using cutting-edge tools such as LangChain, LangGraph, Neo4j (knowledge graph databases).
- Integrated LLMs (Large Language Models) from OpenAI, Ollama, and **Hugging Face** to enhance document understanding and retrieval accuracy.
- Designed and implemented a **Translator Module** to translate documents between English and other languages using Google Translate API and Helsinki Transformer models, while ensuring the original document formatting is retained.
- Leveraged knowledge graphs to create contextualized, query-ready datasets for improving decision-making processes in the medical domain.
- Ensured high-performance NLP workflows by optimizing pipelines for scalability and accuracy in handling diverse document formats and languages.

Client - P&G(Procter & Gamble) - (Role – Senior Data Scientist, Developer)

- Technical Stack - **Python, Pandas, NumPy, MS-SQL 2022, JSON, VS Code, Azure, Azure Pipelines, Docker, GitHub**.
- Proficient in Python programming language for data manipulation and analysis specifically to write code for the data engineering pipelines.
- Experienced in utilizing **Pandas and NumPy** libraries for data processing and analytic tasks.
- Strong understanding of **data structures and algorithms**, with the ability to optimize **code performance and efficiency**.
- Implementation of Data engineering pipeline and implementing unit testing to ensure code **reliability and robustness**.
- Actively engaged in **client communication** to gather data and project requirements, ensuring alignment with business objectives.
- **Adapted system design and architecture** to accommodate client requirements, enhancing overall project effectiveness and client satisfaction.
- **Adapted the system architecture** to accommodate new data sources and integrate with existing client applications.
- Experienced in setting up and configuring Azure pipelines for **continuous integration and deployment (CI/CD)** of data engineering pipelines.
- Implemented data engineering pipelines for efficient data processing and analysis.
- **Partnered with** clients to understand their data needs and requirements.
- **Communicated effectively** to gather and clarify data requirements.
- **Ensured project alignment** through close collaboration with clients.
- Experienced in creating and altering tables for **code optimization**, ensuring efficient data storage and retrieval in MS-SQL Server 2022.
- Populating manufacturing line metrics to Grafana Dashboard
- Designed and implemented ETL pipelines to extract data from multiple sources historians and CCCR DB.
- Transformed large datasets using Python (Pandas, NumPy) and SQL to ensure data quality and integrity.
- Developed mathematical models from scratch for contextualization, leveraging advanced statistical techniques and algorithms.
- Performed complex data analysis and mathematical calculations to derive insights and patterns in large-scale datasets. Applied optimization techniques to improve the accuracy and efficiency of contextualization processes.
- Delivered data pipelines for the purpose of leakage and defect detection and insights for avoidance.

Key Achievement - With accommodating data from multiple sources makes it easy to contextualize data for easy analysis purposes, which improves the manufacturing lines are more adhered to not making defects/leakages in product manufacturing. The defects/leakages have been reduced to half of the previous defects identified.

Key Strengths:

- Proficient in Python programming for data engineering pipelines.
- Strong understanding of data structures and algorithms.
- Experienced in Azure pipeline setup and configuration for CI/CD.
- Collaborative approach with a focus on client satisfaction.
- Proven track record of delivering measurable results through data-driven solutions.

Data Engineer in Larsen and Toubro Technological Services worked in Oil & Gas Domain

Jan 2020 – Oct 2021

(Role – Data Engineer/Scientist)

Remote

(Python, NumPy, Pandas, SciPy, MS SQL Server 2017, Oracle 11G and Oracle 10G, xlrd, xlwt, openpyxl (Anaconda Navigator Framework)).

Plant Portal Data Analysis [Covestro, IPEP (Data analytics team)]

- Implementation for the process of Analysis (Deduplication, Data Segregation, File Reference Missing in SmartPlant portal, finding different equipment properties in plant portal, Different levels of data missing. Different file formats, Segregation and separation of different language formats, metadata, and migrated documents in smart plant portal).
- Used Task, Thread and Parallel process and optimized codes with Singleton and Factory Design pattern. Excel dynamic read and write with formatting using closedXML, OpenXML, EPPlus, Excel Data Reader.
- **Key Achievement** – Computation has reduced time significantly for a large dataset and customer delivery satisfaction was achieved on-time.

SPI (SmartPlant Integration) and SPEC [Covestro, IPEP (Data analytics team)]

- Data Comparison of migrating data from oracle to SQL Server. Generation of Excel Data and its comparison using EPPlus and generating reports. Created Custom controls and User controls, Delegation of custom and user controls, Background task and multithreading with parameterized arguments, async and await tasks, a comparison using multi and different primary keys also with dynamic and different key columns. Data view sync. Comparison checks for BLOB data and other Hash File data.
- Implemented Task, Thread and Parallel process and optimized codes with Singleton and Factory Design pattern. Excel dynamic read and write with formatting using closedXML, EPPlus, ExcelDataReader and Dataset in Visual Studio. Logging facility in python and used data frames.
- **Key Achievement** – With multithreading and parallel processing output time has been significantly improved by more than 90%. Worked closely with cross-functional teams including plant engineers, data owners and site engineers to design, build and deliver end-to-end solutions to their individual data conversion and wrangling problems.

Data Engineer in Quantum Mutual Fund worked in Financial Domain

Aug 2018 - Jan 2020

(Role –Data Scientist/Engineer)

Mumbai

(Python 3.2, NumPy, Pandas, SciPy, ASP.NET MVC, C#, jQuery and MSSQL 2015, SOAPUI AND Postman and Fiddler).

Invest Online (INON) - Website

- Developed web pages with business logic for investment in Q-MF & transaction, Re-Purchase, Redemption, Switch, New Purchase, STP, SWP, SIP.
- ### **Ekyc-CMS, Q-Invest Mobile restful Web APIs, and Window Services**
- Optimized backend SQL code for Investors, used to do KYC according to their citizenship with different criteria & process form in different queues.
 - Scratch development of Rest APIs for consumption in Android app and IOS app.
 - Designed and developed Invest Online (INON) web platform for managing investment operations including Q-MF transactions, Re-Purchase, Redemption, Switch, New Purchase, STP, SWP, and SIP processes.
 - Built Ekyc-CMS system for investor identity verification, adhering to country-specific citizenship criteria and automated queue-based processing.
 - Developed and optimized backend SQL code to enhance performance and ensure seamless handling of investor data.
 - Created RESTful Web APIs for Q-Invest Mobile application, enabling integration with Android and iOS platforms for real-time investment management.
 - Designed and implemented Windows Services to support asynchronous operations and backend workflows.
 - Utilized tools like Postman, SOAPUI, and Fiddler for API testing and debugging, ensuring robust application performance.

Software Engineer in Smart Solutions worked in CRM, FMCG Domain

Oct 2013 - July 2018

(Role – Full Stack Developer)

Mumbai

Inventory management and services - Website

- Capturing, managing, and maintaining inventories and managing. Assigning blocks and holding inventories.

Graduate Engineer Trainee in Godrej & Boyce worked in CRM Domain

Oct 2012 - Sept 2013

(Role – Full Stack Developer)

Mumbai

Timespan- Attendance and Leave management system, Assets management system & Drawer depository safe - Website

- Capturing, managing, and maintaining attendance and leaves for the employee in the company. Migrating assets and system information to standard format.
- Developed and deployed web-based systems including Attendance and Leave Management, Asset Management, and Drawer Depository Safe websites.
- Automated the tracking, recording, and maintenance of employee attendance and leave data for enhanced HR efficiency.
- Migrated asset and system information to standardized formats, ensuring data consistency across platforms.
- Designed and implemented webpages using HTML, CSS, and JavaScript, and optimized backend data management with SQL queries.

SELECTED ACADEMIC PROJECTS

Aug 2016 - Jun 2018

(Python 3, Pandas, NumPy, SciPy, MS SQL, PyCharm Community edition, MongoDB (NoSQL))

Classification of topics with words combination using LDA (Latent Dirichlet Allocation) and Machine Learning

- Classification decides the sentiments of the sentences either positive, negative, or neutral.

CERTIFICATION AND LINKS

- Coding- [linkedin.com/in/krishnabhunia](https://www.linkedin.com/in/krishnabhunia) | [hackerrank.com/krishnabhunia](https://www.hackerrank.com/krishnabhunia) | github.com/krishnabhunia | leetcode.com/krishnabhunia
- LinkedIn & Pluralsight Certificates - **#Core Python 3.6: Getting Started** **#Fundamental Cloud Concepts for AWS** **#Using Python with Excel** **#Data Wrangling with Pandas for Machine Learning Engineers**